

ActInf Livestream #005.1: "Multiscale integration: beyond internalism and externalism" (2019)

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hello and welcome everyone this is the active inference live stream it is september 29th 2020 and this is our episode 5.1 welcome to team com whether you are a first-time listener participant or whether this is not your first time we are an experiment in online team communication and learning related to active inference you can find us on twitter at inferenceactive you can find us on email activeinference gmail.com through our public key based team or through our youtube channel this is a recorded and archived live stream so please provide us with feedback so that we can improve our work all backgrounds and perspectives are welcome here we really want to hear about everyone's individual experience and also as far as video etiquette goes please mute if it's noisy in your background and remember to raise your hand it's awesome that we have so many participants today and so to hear from everybody let's just make sure to raise hand and all keep um a cue of people who have raised their hand so today in act m stream 5.1 the roadmap for our discussion over the next hour and a half will be as follows first we'll have a warm-up section where we give a quick intro and check-in then we'll be discussing the paper we'll talk about the goal of the paper we'll talk about the road map or the section headers go through the abstract and then go through all the figures we'll also be able to touch on some background concepts if we have time and address some excellent questions submitted by sasha and other participants and next week on 5.2 we're going to have further discussion on this same paper so please remember to save and submit any questions that you might have that don't get addressed in this first discussion awesome so here we go into the introduction section please introduce yourself and your location and just say hello to everyone in a short introduction and then pass to somebody who you haven't heard introduced themselves yet so i'm daniel friedman i'm in daviess california and i would like to pass it first to alejandra you're muted yeah well hello everybody um well i'm in mexico uh specifically in guadalajara and this state where i'm living now um but well i mean my professional i'm i'm a full-time associate professor in cognitive and

behavioral sciences at the school of psychology at the unam and also i'm an active member of the community robotics lab at the wyom mexico also and well my research focuses on perceptual optimization processes such as sensory acceleration and sensory enhancement and also the computational modeling of perception cognition and i'm like uh hooked with the predictive processing quality coding and energy principles since long time ago uh it's like um my passion um and i'm like giving a full time some some of my days to the breathing and trying to understand um this this particular free energy principle so yeah that's me let's pass to alex hello everybody yeah my name is alex i'm from moscow russia i'm a researcher in system management school and trying to find a way how to join concepts from system engineering and active inference frameworks so let's say for sasha hi thank you my name is sasha i'm a neuroscience graduate student at davis and my background in a free energy principle is just reading and enthusiasm for figuring out how to apply this to my own work with developmental neuroscience and just looking forward to the conversation i'm going to pass it to mel hello yep can you hear me awesome perfect hi i'm mel andrews i'm in cincinnati ohio right now um i have a background in cognitive science and i've always been fascinated by issues in theoretical biology and switched to philosophy for my phd to really focus on um sort of what are called um life-mind continuity theories and in particular i'm focusing on the free energy principle as a formal articulation of a like-minded movie oh i'm gonna pass this on to richard thanks uh so my name is richard niemeyer i'm an assistant professor of sociology in denver colorado in the united states um my interest is mainly in understanding how we can integrate neuroscience cognitive psychology and sociology into more integrated framework um most of psychological or sorry sociological and criminal chronological theories drawing on a lot of behaviorism from the 70s and 80s it's kind of its theory of mind and how the mind society relates to one another so i'm trying to figure out more up-to-date ways to integrate mind-body and society most of the mathematics that i'm familiar with in terms of kind of how to do that is a lot of fractal geometry graph theory dynamical systems theory which is really fascinating but i've been really interested in kind of the free energy principle it seems to do a much more kind of intuitive fluid dynamic way of integrating different levels of analysis and integrating specifically cognitive emotion action things like that which are really critical when you get into kind of that social level of reality so i've tried my best to work my way through a lot of papers by myself and it's been a struggle so to find a community who's kind of doing this together and can kind of listen to other people talk about these ideas is is really really helpful so i really appreciate the opportunity to be here i'm gonna pass it on to muddy oh classic it was

on me hi guys my name is muddy so i was a i was a phd student with carl briston graduating back in 2016 but i also was a medical student on the md phd pathway so my interest was in applying free energy principle to study stroke dynamics non-invasively using neuro imaging i since left the the medical world i practiced medicine about a week and then uh went into the tech world i worked for google deep mind and started the medical tech company which was acquired by the company i currently work for now versus where i'm i'm the director of ai and we're really interested in spatial web technologies spatial web protocols and building provenance ais that are not only explainable but action-based and adaptable so that's kind of like that and um i will pass it on to professor cliffinger who is quietly set there thank you honey uh hi everybody my name is john clevenger um i am now well i'm in northern new hampshire jefferson new hampshire farm but actually i'm generally out of cambridge mass and um with the mit media lab the city science group and i've been at mit in the human dynamics group for a number of years um my interest actually uh i'm a full convert recent convert to pristin la the the first and cult as they were but uh continued reading the papers that are coming out the productivity of this and working uh with maxwell and casper about how to apply this to uh modeling um modeling new concept of firm new concept the market new concept of community um so we have some projects that we're focusing on that and actually collaborating with muddy on that as well cool let's go to shannon hi i'm shannon crooks i'm in south dakota currently but i'm usually based at the university of california in merced um and i'm in a sensory motor neuroscience lab i investigate the neural dynamics of music cognition and specifically how we process rhythm and then interested in how predictive processing approaches can help us understand how we perceive rhythm and music i'm also interested in how groups of people interact when they're experiencing music when they're playing it or listening to it and in larger dynamics of crowds i'm excited for today's discussion um specifically for how you know a crowd could have a different markov blanket based on its interactions throughout maybe a musical performance or a sporting event cool i think we have maxwell muted maxwell sorry about that hey i'm maxwell ramstead i'm a postdoctoral fellow at mcgill university and i'm one of the main architects of the multiscale active inference formulation that we'll be discussing today and the first author on this paper multi-scale integration beyond externalism and beyond internalism and externalism and i will pass it off to alex kiefer uh hi everyone i'm alex um as nestle just said um i'm coming from a philosophy and cognitive science background um i got my phd at the cuny graduate center um currently um at monash university in melbourne but not actually there because of the pandemic so i'm in

the new york city area um and i'm interested in uh i'm coming from sort of a traditional philosophy of my perspective i increasingly got into like connectionism machine learning modeling eventually the fep and i'm interested in how this stuff ties into fairly traditional representationalism on the one hand and sort of physics on the other and how these come together so that's my angle and this and i'm really excited to talk about this paper cool i think stephen and then if daniela jumps back in we'll go to her but stephen i think you're the last one here oh okay hi i'm stephen civit i'm in toronto and i do community development projects using theater and immersive participatory approaches and ended up developing a process called social drama topography in south africa in 2006 and trying to scale that process in different ways has looked at how embodied cognition fits into this kind of process of engaging communities to make embodied maps together around issues and that's kind of translated over time into other sort of immersive practices and i was looking at embodied cognition over a few years and i kept finally seeing this work with carl frist and i thought oh my goodness i've already gone so far down all these avenues and i've seen this other thing and then suddenly i found a page which spoke to how many citations he had and i thought wait a sec and i suddenly saw that his work was actually bringing all these different threads together into this unified framework so yeah i'm really excited by that awesome well thanks so much everyone for coming out to participate um it's really a spread of countries and levels of experience and fields of expertise so it's about bringing it all together and coming together around these topics so just as a warm up and to hear from everyone who would like to speak by raising their hand let's just start with asking what drew you to this paper or topic today so all star and while other people are raising their hand and i described this a little bit in the 5.0 video but i think the idea of integrating multi-scale systems is really um interesting and there's so many ways to go about it and it's almost undeniable that we're living in multi-scale systems but the question is what now and so i'm always interested to hear about systems for thinking about what now and also for different people's perspective great muddy and anyone else can raise their hands muted though with a video off it's the fine line between muting the background noise and being ready so i understand but go ahead muddy my apologies yeah i think um nested systems are an incredibly important way to to view real dynamical systems that we see all the time in the spatial web so we constantly have streams of data coming from iot and then we're kind of met with the scalability challenge of how do we actually model a system accurately to represent the dynamics that we that we wish without getting bogged down in this sort of field of complexity so by abstracting away these these layers into really really meaningful nested

heterarchies is is really really key in being able to practically deploy some of these you know novel learning paradigms in real world spaces so i think without this sort of quote-unquote dimensionality reduction i mean that in more of a philosophical sense than a mathematical sense we're going to struggle to apply ai at scale so that's why i'm i'm particularly interested in the work that maxwell and the free energy principle group is doing cool let's hear from maxwell then sasha and then i also see other people but just those two next maxwell also muted sasha do you want to go first i don't i don't have to go next necessarily sure i'll uh briefly say a bit um i was drawn to this paper because as a neuroscientist i think a lot about multi-scale um levels and to decide at which level a system should be studied because even though um we're studying something at a molecular level or even in a different species we ultimately want to make claims about larger cognitive processes and how to do that without abstracting away really important aspects of the system at the scale that you're studying it it's a huge question so um i really like thinking about these markov blankets at different levels and where to draw the line cool maxwell then richard then alex k i think my remarks are gonna echo a lot of what's been said um i got into the free energy principle stuff originally uh because i was looking for a theory that could address dynamics at and across multiple different uh well the way i used to see it is ontological levels uh in working with the free energy principle i've come to reject the traditional metaphysics of uh of ontological levels uh in favor of uh you know this view that there are big things and there are small things and uh big things are made up by small things and big things tend to be slow and small things tend to be fast and what you actually have is like this nested series of systems that are made up of increasingly big things um and yeah back when i was starting up my phd in 2015 uh there wasn't any theory that i could see that was really able to not only define dynamics at each of these different levels but propose a way a mechanism by which they were all integrated and this is what i found uh via the free energy principle so i i sort of made it my thing uh from 2016 to now actually to explore the multi-scale uh implications of this framework awesome richard then alex k yeah i think um one thing that i've been really drawn to the last couple of years is the mechanistic philosophical perspective of of carl craver and lindsey darden they give you this really nice i think tool set on how to talk about decomposing mechanisms levels of mechanisms which is very intuitive and i think works well with with sociology and biology which has this very clear conceptual way to move up and down quote unquote levels of reality but the thing that's i think always been missing is how do i take that conceptual language and match it with an empirical framework that allows me to um model those levels of reality in a

way without losing any of the subtleties any of the the conceptual apparatus that is so critical to talking about those those different frameworks and the more and more i get into the free energy principle it seems like it can give you that empirical that has a very nice natural transition into that conceptual language i'm so used to that gives you all of the stuff you're trying to talk about conceptually but then also gives you these empirical tools that allows you to go above that and talk about the organization of the mechanism in a very nice precise empirical language um that's just impossible to do i think with a lot of existing methods in the fields that i'm used to cool let's do alex k and steven and then i'm gonna go to the second question um so i i just recently sort of got into this whole multi-scale picture um associated with the fpp and um i guess my interest in this is um it seems to me like like i'm i'm nearly 100 convinced that this is a really good description of reality at multiple scales but i'm not completely convinced that this has the radical kind of philosophical implications that that some people think it does i mean this paper actually strikes kind of a nice middle ground but um i guess to put it briefly like i view this as we've got nested multiple systems each of which possibly has a physical as well as a cognitive description and maybe the bound the bounds of cognition within that hierarchy are maybe less strict than may have been thought like i'm happy describing cognition or ascribing cognition to systems at various levels within this hierarchy but but i don't see that it's so hard to pin the cognitive properties of one particular system um onto a particular physical system so i'm not i'm not sure anyway i hope that that gets we can get through some of that also i'm not sure which system i am in this hierarchy of embedded systems that you know my brain involved many systems in itself and that's self-embedded in larger systems so i'm trying to figure out which type of system within that hierarchy an individual person might be identified with cool stephen and then we're going to go to the next question to muddy and mel or anyone else who raises their hand yeah i'm very interested in how this gives a kind of action orientated miso level approach which is really useful because when i'm working with say a community trying to understand how to connect to the individual to the group to the team to the organization or maybe influencing change at the municipality or broader social levels there's this real problem with bridging scale um and it ends up getting caught up in all this languaging that whichever field has been you know the psychological perspective or this perspective on that perspective so this gives a kind of seems like a very neutral clean um way of understanding how people make sense of different levels and that's really valuable i think that's been missing from community psychology is really actually psychology itself suffering from that because it's kind of stuck on this level of um

perspective or interpretation so this gives a way to actually go into the sub-paragraph so for me getting into all of this has actually opened up all this sub personal neuroscience site which i actually haven't really got into before so it's kind of interesting nice thanks for all these responses also raised her hand yes so next on the stack is muddy and then mel and feel free to answer either the first question or the second question about what a multi-scale system you work with is so muddy first um great so they actually kind of uh fall into each other so one of the things that i wanted to mention was um the practical challenges of implementing um inference in an industry right which we kind of do it in a black box but i think it was sasha who mentioned that or alex who mentioned he had an interest in this as well um and what we find is that the the practical correlates uh in terms of hardware and chip design that we need to to deploy this stuff in reality is is difficult based on the data sets that we collect so one of the systems that uh we're currently modeling right now is warehouse pick and pack working for for amazon and understanding that the the systems that drive why people pick certain boxes from certain shelves for example is like a multi-nested system of everything from how valuable an item is to how tired the pick and pack worker is even down to the function of how tall they are versus how high on the shelf the particular item on the waiting list is so understanding how to pick that apart and um deliver value essentially in industry eventually so yes we have this wonderful way of viewing the world but we also have a practical set of tools which are a wonderful way of modeling the world as well and so having this nested system i think is really really important in terms of biting into the problem of practical deployment of these these types of models at scale in order to deliver value now i don't just mean business value i mean you know inside value as well so those are just a couple of comments that i wanted to to introduce into the discussion as well so thanks for that mel and then anyone else can raise their hand hi um i was drawn to this topic because um i've been trying to convince maxwell for a long time that the fep is just um that basically we should we should be taking as instrumental as an approach to this as as we possibly can this is like the bare bones uh it's it's just a model it doesn't constitute knowledge of the world it doesn't constitute falsifiable claims it doesn't constitute uh et cetera et cetera et cetera it's just a model and we just we use it to get at things in the world and and and that's it um and this paper's uh interesting to me because in it maxwell is the one pushing the instrumentalist line that that i don't want to bite uh namely that's a that there's nothing more to the boundaries of systems than is it useful to model it that way right is it is it can it can it be usefully modeled as the battery is here for me i'm i'm very um i'm very willing to privilege the organism as a privileged scale or level or unit of analysis

and i'm very willing to privilege the brain as well so in that order interesting fighting words mel um shannon and then anyone else can raise their hand too i i feel partial to those those last words from mel on privileging the organism or the brain um but that could be just because i i study the brain i want to know the the neural mechanisms that are giving us any of our perceptions or engagement with the world um but i think i was drawn to this topic because um i was drawn through predictive processing literature and like um andy clark's work and then eventually on here to the free energy um but a lot of what was written and like is now has these great mathematical models um that we've touched on before like an ability to translate um maybe some sort of phenomenological experience into a formal model um it meshed really with my studies in music and what like teachers of music will instill upon you in that you're basically through your musical experience you're modeling something about the human um so you're choreographing aspects of tension and release which sounds a little bit like balancing um predictions and prediction errors um and you're trying to maintain a certain boundary within a musical piece of the emotion that you are experiencing or that you want your audience to experience and this sounds a lot like a boundary on a certain phenotype of experience um and just sort of this like superficial matching of the language of musical expression and then this empirical um like statistical language is sort of what drew me in and i suppose that answers the multi-scale system i work with um music and musicians and people cool very interesting we'll go to richard and then the third question yeah kind of building a little bit on on mel's point in terms of the multi-scale system so obviously as a sociologist i deal with groups i deal with society and you know i can take an individual and i can take two individuals now i have a dyad i have another one i have a triad i can model that as a network i can collapse the network down and model it as a dynamical system and now i can talk about organizations i have multiple dynamical systems talk about them interacting now i have a network of coupled oscillators i can go in i can use fractal geometry to talk about how properties are scaling up and down from the dyad the triad all the way up to that network of coupled oscillators and at the end of the day i i'm never really confused about whether or not i'm really talking about networks or am i really talking about dynamical systems they're all just models that allow me to capture the phenomenon that i do believe is real i do believe society exists if you ever want to get into a very strange conversation go to a conference of sociologists arguing about whether or not the united states is real and my comeback is always don't pay your taxes on april 15th and you'll figure out very quickly if the united states is real or not and so i think you know a lot of these philosophical questions are always

fascinating especially at three o'clock in the morning in a pub when you've had too many to drink but i think ultimately in the day you know there is a concrete reality the question is what empirical model just allows you to model it as true to the theory or the concepts that you're that you're dealing with um without losing information you know unnecessarily so i think you know again i think any empirical model that allows me to do that is fascinating um but at the end of the day i do believe you know the humans that i'm ultimately talking about are concrete and are are real nice i'm gonna introduce the third question we'll go to buddy first here and anyone else can raise their hand the third question is why does our philosophy of systems matter um that is quite bizarrely scary and psychic because that's what the point i was going to make just to just to piggyback on on richard's comments just which is um i remember these discussions these really really lively discussions about you know how representative is this model of reality and i remember at some point just thinking well how useful is this a question to ask you know at some point is is the model just a tool a means to an end and do we employ bounds on what we expect from these models in terms of their utility and if these models and their outputs satisfy these bounds great you know i i sometimes i i see some of the the discourse around free energy principles and and what it means philosophically and we lose what it gives us in terms of explainability and an actual measurable dynamics so i actually fall probably like very far on the side i don't think our philosophy of systems matter at all i think our explainability empirically measured data is by far the most important thing and trying to explain systems where our data sets are incredibly sparse so for example the entire hierarchical organization of the brain from cell to effective connectivity i think is um sometimes missing the point of just how valuable these novel uh schemes can be so that that's kind of what i wanted to touch on that nice stephen and then anyone else before we close out this question yeah i think that that empirical pieces and practicality or practice application is really a good point one thing that has caught me out though and i've realized is is that perspectivism is very much part of systems generally and it's also part of psychology like we take a perspective on stuff and there's this assumption that everything well what's your perspective that word just gets used all the time and it's like but the thing is that the free energy principle gives you a way and then they say okay and now we're going to do stuff that's performative or arts based and it's a bit like we don't know what to do with that so we just call it another field right and then they do that but the the the the active inference gives you realize well there's there's a way that you can abductively or from within be creating meaning um during experience and it's um i suppose a bit like that person picking stuff from the shelves and they're not sitting there as

they do it having a perspective like they may take a perspective if you give them a qualitative survey later which is what nearly all social science does so you take us perspective a first-person perspective on your experience but the experience wasn't first-person perspective the experience was just the experience so this gives a way for that to all be kind of um unified in some way cool let's go to alex and then richard then close out this check-in um just really briefly i mean i think i think it matters philosophy systems matters because i think the philosophy well i guess i'm sort of a holist in a way at least in terms of explanation so i think it matters um the philosophy of every little bit matters uh because it connects to everything else but also this is sort of a question can be applied at multiple scales so so it could potentially matter a lot i guess i mean that doesn't really answer the question of why philosophy matters ever but i think it's because it's part of a big theoretical framework i'm sort of acquiring about this stuff so these questions about realism and instrumentalism don't land all that well with me because i mean for example i think the fep if it ends up being part of our best overall sort of account of things then i think it's confirmed or disconfirmed by proxy whenever anything any observation is recorded right so um unless you want to say that it doesn't have any inferential connections to anything else in which case it's really weird i don't know why we're talking about it um anyway that's enough for me perfect richard and mel to close out the section thanks yeah i think in many ways and this is probably just maybe disciplinary specific but the philosophy of systems and the philosophy of what we do in science i think is incredibly important in the social sciences because so much so much of our concept so much of our theory that that we deal with now is is deeply indebted to dewey and uh conrad and you know hegel and these these uh philosophical arguments about the relationship between mind body and society what is a thing what are the boundaries of things and these things have been baked into our theories and many times we don't recognize that they're there and they're they're deeply affecting concepts and how things are operationalized and what we consider to be a truth value or not and to go in and unpack those theories find those philosophical arguments and then interrogate them from these kind of emerging perspectives and say you know was whitehead onto something about kind of you know uh the extension of reality as an ontological category or was he not and if he wasn't and this new principle can deal with a lot of those ideas in ways that were significantly shaped uh sociology then great we have something to kind of move on to um so i think that philosophy is always this thing as much as i try and get rid of it it's always coming in the back door some somehow some way and i think just being sensitive to it being comfortable dealing with it and interrogating it from these new

perspectives i think is always going to be something that's very important because it's always there in the background somewhere if you dig deep enough thanks well said and then mel and then we'll let maxwell have a last stop as well in my mind philosophy doesn't answer questions it raises questions um and the empirical sciences attempt to answer those questions but you need philosophy but don't chew on the computer jesus christ you sorry you need philosophy to both raise those questions and then to uh to tell you how those questions have been potentially answered by the empirical results so um there is no such thing as as kind of science operating in a in a vacuum without philosophy of science that's my opinion uh that that there's this necessary dialectic between the philosophy of science and science itself um going on cool agreed maxwell and then professor clippenger if you'd like um well so i think that there's a there's a methodological component to this as well i i want to echo what what everyone said earlier uh you know if if we're gonna study uh multi-scale systems it's good to get our ontology straight and you know the fact that nature seems to have this multi-scale structure suggests that our theoretical accounts of nature should also reflect its uh multi-scale structure but i think from a more pragmatic point of view um i i like to think that a lot of the uh kind of single discipline uh topics have uh been covered in in a very significant way but that there isn't there isn't the same kind of um you know established work connecting different methodologies that operate uh at different scales so i think that you know the getting this systems take right not only has theoretical implications but also has practical implications uh for the way that we conduct uh research namely that we should aim to form uh multi-disciplinary groups that address questions of interest from several different angles simultaneously so and this is echoed in the paper that we'll be discussing today where what we're essentially trying to do is advance uh on the one hand a pluralistic ontology a multi-scale uh ontology and on the other hand uh a a uh methodological pluralism uh where several different uh sets of methodologies can come and interact together nice professor cliffinger or anything else there you're muted but yes um okay i have to apologize because i have to leave early but this is a great discussion a great beginning i'd have to i just want to throw my hat in this i i'm sort of in in a muddy's camp uh and looking at explanatory power um and i think a lot of that is there i i'm so sort of uh ditto with with what maxwell is saying in terms of providing an integrated framework uh i i of course as a philosophical the philosophy has a place in it but i think we're there there were so many losses embedded in historical situations that are really not that i think we're at another level of understanding that it's what becomes really important is being able to start to test these things and be able to rigorously test them and because there are a lot of very powerful claims

that are being made that if they're true they're quite transformative but they need to be tested awesome thank you everyone for this very exciting check-in and i'll just add one comment on why the philosophy of systems matter and it's examples that are brought up by um helen longino in her book studying human behavior uh which level of analysis which discipline we default to molecular neuroscience or environmental studies if we default to a higher level of organization or lower level to look for our explanations philosophically it's going to directly translate to what kinds of interventions we think are going to be possible in the system and so if the story about for example aggression and sexuality which is what professor longinus studies in that book if the story is it comes from jeans well then you're going to be less likely to undertake affordances related to the environment if the story is um about the environment you're going to be unlikely to take affordances related to molecular interventions so even before we say what it actually is and that's of course a complex question or what is the best thing for the system i i would fall on the side that it really does matter in the background and sometimes it's just the water we swim in how we think about these multi-level systems so let's just like for a three second aside i i will happily admit that i'm a staunch empiricist somebody who worked in industry basically the moment i could and so i you know what i've i'm just conceding a little bit that actually you're right without the philosophy of science we don't introduce novel ideas that are worth testing in the first place right and that discourse is what leads to the models that allow me to try to issue value from that quote-unquote value from that so i am backing down just a little bit from my empiricism after this discussion so i just wanted to say that yeah nice a live conversion story you heard it here first but even pragmatism is defined within a use view so it's like all models are wrong some are useful and then the second part of that sentence might be useful for whom or under what conception there's no model that's just useful outside of a value set so pragmatics isn't just like let's just do the most useful thing there's not just one useful thing so after that extremely interesting check-in let's get to what the paper is and what the goals of the paper are so the paper is multi-scale integration beyond internalism and externalism by ramstead kirchoff constant and fristen from 2019 and what the paper writes their goal as is here we focus on making sorry someone needs to mute um i'm going to mute that person here we focus on making explicit a description of the boundaries of cognitive systems that we think follow from taking seriously the inactive embodied and extended nature of cognition this is the idea that the boundaries of cognitive systems are nested and multiple and that with respect to its study cognition has no fixed or essential boundaries and so it's all about this taking it seriously with

a multi-scale integration so how are we going to take the seriously i'm going to run quickly through the section headers and anyone can just raise their hand and i'll pause the just to address the question so first there's an introduction of the paper which makes sense because there's so many areas that are being tied together here there's the first figure which is about markov blankets and active inference in the second section of the paper which is called a variational principle for living systems first they talk about this variational free energy formulation for living systems and show a figure of how the free energy principle and self-evidencing are related to living systems they then talk about generative models and action policies because when we're talking about internal models or internal states of an organism we're talking about generative models and action policies they then talk a little bit more specifically in the text about markov blankets and the boundaries of cognitive systems which recalling their goals they're going to take us to somewhere where these boundaries are real and they're not equally prescient as they say however that doesn't imply that there's one single correct level of analysis this is related as maxwell also brought up earlier on the formal ontology for the boundaries of cognitive systems because the modeling approach the descriptive or action-oriented modeling approach that we take to a system is related to how we think about it and so that's the background if your ontology has a certain type of scheme pre-built pre-installed then your models are downstream of that so when we're talking about philosophy of science philosophy of multi-scale complex systems we're talking about what ontology will let us draw the boundaries around these systems in a way that's yes useful but also aware about these other aspects and the fact that we're not the only perspective on use out there this takes us to section three on cognitive boundaries specifically this debate long-standing between internalism and externalism it ends with an ism so it's a dogmatic belief and it um is just what it seems to be it's about whether the external factors in cognition or whether the internal factors in cognition play a more relevant role or a primary role and the internal and the external are implicitly at the level of the organism and so one of the i think strongest stances of this multi-scale analysis is well isn't there an internal and external to the group isn't there an internal and an external to the enacted ecological cognition like a human in a building with a computer what does internalism and externalism mean when we're talking about niche construction when we're talking about sociological systems and so to explore that first they talk externalism in the context of radical views of cognition so radical means from the root and so to take it from the root that externalism is the nature of cognition would be to kind of have a causal model where the outside things are more causative going in rather than the other way

around in figure 3 there's a very nice abstract representation of operational closure which is actually related to kant and the idea of teleology and whether the organism is the fundamental unit of analysis in biology or whether other levels might be important too and figure four shows from an autopoietic point of view about autocatalytic closure and so it's relating using a few different metaphors including uh autocatalysis like a metabolic loop metaphor for closure as well as an informational closure that will be returned to when talking about the markov blanket internalism is then given their say and it is pushing back against externalism if one can imagine what radical externalism looks like you can probably imagine what radical internalism looks like and then after bringing up in section two the variational free energy principle and then in section three framing this debate about internalism and externalism in section four we combine sections two and three which is to say that we apply the free energy principle formalism to taking it seriously integrating multi-level systems between internalism and externalism or moving beyond it and that implies re-analyzing generative models what they are and how they're used to study cognition a section called inactivism 2.0 which is a callback to earlier uh generations of an activism then there's a section on nestedness or how to study cognition beyond the brain so okay neuroscience or behavioral neuro is about what's happening in the brain but that's nested within a body that's nested within a social group so how does this nestedness change how we study cognition in the brain and beyond and that's represented in figure 5 with this blanket of blanket ideas and in figure 6 with a multi-scale self-organization and active inference figure and then the paper closes with a movement towards multi-disciplinary research heuristics for cognitive science so just as far as today's discussion goes in the next 30 minutes we're gonna try to run through the abstract and the figures and then we'll have some closing thoughts and then that will hopefully get everyone excited and interesting thinking about different types of questions which we'll be able to address in live session 5.2 in the service of the half hour we have to get through the six figures i'm just gonna put the abstract part one up here what they're doing is presenting a interpretation of the boundaries of cognitive systems and they're wanting this to be a corrective for the debate between internalist and externalist positions the second part of the abstract they discuss how they first survey key principles of this radical extended or externalized cognitive perspective also implicitly in the context of the internalist perspective and then they develop a multi-scale account building on the free energy trying to move beyond this internalism externalism debate and then in the third part of the abstract they close up that discussion by saying that their approach does not privilege any specific given boundary of the

world but it also doesn't argue that all boundaries are equally prescient so that's going to be something interesting to unpack and then they close just by saying that they are drawing the boundaries of cognitive using this multidisciplinary research approach so let's get to the figures and this first discussion let's just try to understand what are the figures about what is being shown in the figure just a first pass level visually what do you see but also where does it fit a little bit more broadly into the a to z of the paper or just what did you think about the figure what did you see in the figure or what were you wondering about so here we are in figure one would anyone like to take a first pass at describing figure one what they see or um what it shows or means for the paper all right so i'll start describing it and someone can okay max well go ahead well so this is uh this is a depiction of a mark called blanket uh and so markov blankets are one of the uh kind of building blocks or ingredients uh that go into uh the free energy formalism um and essentially a mark called blanket uh well mathematically what it does is uh if you have a set of random variables um and you want to effectively uh map uh or individuate is a random variable or a set of random variables from another set of random variables so for instance you know you want to individuate a system like the brain from all the other uh you know dynamical uh systems with which it interacts well what you do is that you stipulatively define uh a third set of states uh between uh the the internal states of the system that you're interested in discussing and the external states of the embedding environment this third set of states is called a markov blanket and it's defined by the absence of connections so effectively uh the internal states are influenced by sensory states but do not influence them and um external states are influenced by active states but do not influence them so by precluding these two uh basically directed edges of this kind of connected graph what we're doing is uh essentially formalizing the kind of channel through which any interactions between uh the environment uh and the uh systemic well the the internal uh states of the system have to have it has to go the interactions have to go through this channel effectively um so uh yeah that's what that's what the formalism does for us it it allows us to individuate a system formally in terms of the interactions that it has with this with its embedding environment great and mel go for it yeah i think a lot of a lot of people publishing on on the fep and the markov blanket formalism but even more so in kind of the discussions you get after a conference people really expect of the markov blanket formalism that it's that it's picking out not um token distinctions but type distinctions so so people really want the markov blanket formalism to cut at natural joints that is to find kind of the type boundaries between natural kinds sorts of natural systems um and it really doesn't do this it really really doesn't do this it really picks out

individual systems that's what it does is it finds token systems um and naturally i think a formal methodology that that isn't about um it's not about finding privileged scales privileged uh spatial scales or privileged time scales it's uh it's about finding particular systems and so i think naturally when we employ this formalism it's not going to lend us a view of um of privileged scales because that's not what formalism does cool and i think that's kind of against goes against what people tend to think the markov blanket formalism is for awesome muddy um i was just gonna say that uh so i would say that most people on this call are far more intelligent than i am and i i try to come up with with my own sort of baby interpretations of this which to my knowledge nobody has been angered by what and i think really what this tells me is if i have an agent that is interacting with an environment the only the only channel by which they can actually update their beliefs in order to change their actions say comes through this sort of border state these sensory states so it's it's interesting in that um i don't consider this well i mean it is a separation obviously between two between two um objectively different realities should we say but it also um it speaks to the the force with which uh um the the analytical link between the two through a certain set of sub states which is which is really important right and this is this border set on on all your sensory states so your sensory inactive state so that's kind of how i've always viewed this i have two sets of realities that are trying to come together in terms of what's going on one of them has a belief state one of them just is and the the belief state is really trying to figure out what is and the only way it can do that is by modifying its sensory state so that's just something which for maybe people who aren't actively part of the community might make it a little bit more reductionist and simple to to digest but if anybody disagrees with me please let me know nice well said um richard and then if anyone else wants to raise yeah so i'm sure this is probably a naive question but you know when i was listening to maxwell talk about it and looking at this figure do you do markov blankets work in the sense of i i hypothesize a priori that there's a thing called an environment there's a thing called an organism there's a thing called sensory states and then the markov blanket allows me to go in and find it statistically and determine whether or not these things can be isolated you know empirically and therefore kind of reinforcing my idea these things actually exist in reality or can i just feed a markov blanket a whole bunch of random data and it's gonna find something you know kind of that garbage in garbage out model so like when you're when we're talking about barcode blankets and finding these boundaries um how exactly does that does that work let's have maxwell just directly respond to that very nice question and then we'll continue on the stack with stephen and then mel uh sure um to answer

that question it would be easier if you could change figures and go to figure let me just check on the in the paper it's figure five sure so this is my favorite figure uh in any of our papers uh and yeah so this this basically tells you uh what what's basically going on here uh in the construction of a markov blanket so um in more recent work this has been formalized in terms of two operators a grouping operator and um a uh a dimensionality reduction operator and effectively what we're doing is well so uh to answer the question richard's question directly uh in principle uh what we've been able to do recently is uh develop techniques that more or less automatically although you know we can discuss there are a few design decisions that have to be made these have been discussed in a few of our discussion channels recently i'm sure mel will uh i'm sure you'll remind me of this uh but basically what happens is um what you do is you start with a bunch of states right so we're working from uh dynamical systems perspective and we define whatever states are of interest for us and essentially the markov blankets are constructed by looking at the relative rates of change of variables with respect to each other so mathematically what we're doing is taking the the partial derivatives of each variable with respect to each other and and then using effectively these rates of change uh to construct the mark all blankets automatically so you start with a bunch of states uh here i'm at the bottom panel in the figure where you have essentially these nine circles that represent states so i mean these can be anything whatsoever uh in a recent paper that was pre-printed this has been used to analyze fmri data so in in that paper each of your states is one 3d voxel um and basically what you do is you write down an adjacency matrix uh so an adjacency matrix is uh effectively a a a matrix where you have all of your states by all of your states and the entries in the matrix are quantifying the effect the influence that one uh state has on another and if you take the if you take the jacobian of this adjacency matrix you're replacing all of these entries with the partial derivatives of one variable with respect to others and uh so having said all that you you can construct the markov blankets analytically by noting the the absence of dependencies in this adjacency matrix so zeros in your jacobian mean that the partial derivative of one variable with respect to that other variable is zero uh which is another way of saying that basically the two variables in question are conditionally independent in that direction um and so i mean there are a few modeling decisions uh that you need to make effectively uh there are most there are two to my knowledge one is basically uh going to be deciding which states are internal states uh effectively so you're going to say okay if this one is an internal state then its blanket is this and the external state's relative to it or that uh that arbitrariness isn't all that arbitrary in the sense that the you know uh the blanket structures are

gonna be essentially the same it's just what what is what's going to count as an internal state might change um and then there are a few design decisions uh when you're when you're implementing this uh basically you have to when you're constructing your adjacency matrix you have to kind of decide what counts as a cutoff for interaction so those two things might make it less objective but in principle if you'll allow for these two kind of modeling decisions you can feed these algorithms like a time series data for example and it'll chunk out the markov blankets from it very nice answer maxwell thank you so we'll go to stephen and then mel then alex k and shannon okay just get my head into it but um one of the things that i think this also is maybe not always easy to imagine is that is the dynamical nature of all of this that it's talking about states so it's not you're never being one thing you're always in this state of um unfolding and that the markov blanket then is that kind of statistical kind of um you could say record keeping at the different scales ultimately i would say everything's at the cell level to some level because that's the kind of engine that's enabling everything else to be created the only thing i kind of feel that is important to maybe distinguish when talking because i think there's a danger with the word systems is that there's like i see these blankets as more like a swarming of um statistics which creates the structures and the networks such as the kind of leafs or whatever and they and it's that kind of structures which actually are kind of in the world so to speak and those structures have systemic properties i.e they become the um support mechanism for a plan the leaf does or but the structure has come out from the swarming of the um actual blankets so that's just i think i'm still trying to get my head around how to pass it nice we can return to that mel and then alex k yeah i think i think maxwell basically said it all but but the the short of it is um you have some time series basically and that could be that could be um real data or that could be uh sim data that could be simulation right and you're pointing the mark off blanket formalism you're directing it to the internal and you're handing it your threshold for for how conditionally dependent should be conditionally independent enough to count um and then that does the rest of the work for you that that that will partition your state so so you give it the internal states and you give it the kind of threshold for conditional independence and that partitions the rest into external states uh sensory states system active states cool alex k then shannon and muddy yeah just a really brief point i'm just wondering if one way of thinking of this is that the markov might get partitioned insofar as its objective um together with some data like maybe it maybe it cuts nature at particular joints but it just doesn't cut at types of joints in a way um so i don't know maybe we can pick that up later well yeah that's a good point if i can just jump in here i think mel is right

when they say that uh like the so it's it's the the markov blankets are always going to pick out individuals uh if there's anything like a species level regularity that we can express or like a you know irregularities that are specific to members of members of the same group uh that'll be more in terms of the generative model non-equilibrium steady state density uh so i mean you know this is also discussed in the paper but essentially what we're saying is you can imagine a joint probability distribution all over all of the variables of the system and implicitly also over some fictive external states that the system is interacting with and this entire joint density uh you can think of as the phenotype of the system uh and it's known as a non-equilibrium steady state density or a generative model so i think yes things that pick out types have more to do with this uh joint density that tells you like about the kind of creature that i am whereas uh as mel was just pointing out like these are uh these these markov boundaries are effectively the boundaries of individuals every time so cool shannon and then muddy i think i might be echoing a lot of what was said already um but on on what the markov boundary is is is separating um it's always going to be a decision on the part of the researcher for for what correlations seem to matter for the explanation that you're looking for for the particular system's behavior so like if there's certain long-range correlations that are constraining short-range correlations whatever your whatever you're wanting to answer so if i want to answer how um a certain musical phrase affects the movement of the person listening to it particular the particular um like enter the particular part of the system that's interesting to me is maybe not going to be the electro electrical firing of a neuron at that moment because the electrical at that really short range correlation that really short-range time scale isn't going to have a huge effect at the long range time scale in my explanation even though it's important you can't have the reaction without the brain having some electrical activity it's still not the most informative scale of analysis nice muddy and then mel awesome i um so again just just bringing this back to uh a level of understanding that allows me to like get some sort of intuition why this hierarchical uh markup framework is is really really important and useful in defining dynamical systems so one example you know my background in in medicine reason why we we were interested in functional neuroimaging is we wanted to understand at what scale can we intervene in certain effective connectivity architectures in order to make clinical difference so understanding the boundary states of which pieces of this puzzle are genuinely boundary stage you know genuinely fall within some markov blanket is really really important and one example of this is if i take my auditory cortex effects and my visual cortex at a hierarchical level of processing in one could say that the internal uh circuitry that allows me to

interpret you know the the uh signals from my cochlear versus my retina are not correlated with each other however at an integrative state above this when i'm trying to actually make some sort of cohesive view of the world they are both of course within that same markov boundary state so here's an example of a hierarchical separation of boundary stage which is really really important not only from an empirical modeling decision in terms of like what level do i abstract but also from an understanding perspective you know one can say assume a boundary state and then test empirically if this if this is correct would say some sort of a dynamic causal model and simply just by changing what you think the actual markup boundary states are or you know the scopic level of that markov boundary um you can start inferring things that are really like clinically useful for so the you know the motor cortex and um the cerebellum quite clearly form a markov boundary for motor processing but then motor cortex and s1 also form a markup boundary in terms of action decision making so this tool it's it's really a beautiful explanation of the world but it's really a beautiful modeling tool as well we're abstracting a way hierarchy called noise in terms of your model so i just wanted to add that in oh i think that's a great point i mean it it shows the kind of meta modeling uh strategy that we have going on here because so uh in another paper that i think you've discussed on this podcast a tale of two densities um uh i would i'm not sure that you did we've discussed a few anyway uh in uh in in that paper uh well actually maybe we should just i'll i'll just seed my my turn sorry no all good all good let's return to mel and then anyone else can raise her hand yeah just to quickly clarify um contrary to what alex said i think um the term the the phrase cutting at natural joints as it's used in philosophy of science actually refers specifically to to type distinction not not token distinction so it's actually it's about finding natural kinds and natural kinds are are definitionally type distinguished and altered distinctions yep also if i could add one note there from my field of gene expression analysis there's often a few different ways to construct a network of relevant genes for you could uh pull down and see which proteins are physically touching or which transcription factors are binding to which genes so that might be a protein-protein interaction network or a gene regulatory network or there are statistical networks like co-expression networks which are very related to things like um grainger causality or effective connectivity they're statistically derived and there might be two elements like two proteins or two genes or something that are correlated in their expression but never touch and those will come up on the functional connectivity matrix um or you might have two proteins that touch and actually are very integra integrally important for function but they're not expressed in a correlated fashion and so that might lead to them being missed on the co-

expression graph or the co-activation graph on the brain and so then somebody might want to say well what's the what's the real map of the territory or what's the real network and that's where this meta modeling question returns and uh related to the carving of the joints yes on the top of figure one we see okay let's carve out the organism from the environment but that's not just a really close line around the organism that actually includes some long-range statistical dependencies as well as things that might be touching but not that important for regulating organismal states so there's multiple perspectives that are required and certainly multiple disciplines that are required when we want to be asking how do we design interventions or given this causal structure the endpoint isn't just scaffolding out this infinitely complicated reference map it's an action-oriented framework that we want to be talking about multi-skill models and then also knowing how to work within them sasha and then anyone else before we go to figure two um great point daniel about um the inter different kinds of interactions that exist in systems and whether it's um statistical or like physical correlation and i think the the kind of overarching theme here is that you wouldn't know unless you measured it you need to know what sorts of variables are important in order to even see statistically if they're going to be related so that is where this kind of framework is really critical because we need to um measure the relevant things in our system and if things are happening at scales that are too long or too short outside the bounds of our measurement then we'll never know nice we'll do maxwell and then stephen and then move to figure two well i just want to pick up a bit on this meta modeling thing that that had been mentioned i think it it's one of the strengths of to it's one of the strengths of the framework to on the one hand um describe like the workings of organisms using this uh free energy minimization stuff so we're sort of uh we're sort of adopting uh the theory itself the free energy principle theory itself adopts an instrumentalist stance on organisms uh to say well what organisms are essentially doing is leveraging the statistical structure of their bodies in movement which we can interpret as a generative model as a joint density over all the system so using the statistical structure of their bodies in movement to generate adaptive patterns of behavior so at that at that first kind of theory level there's an instrumentalism uh that's going on and there's a kind of metal there's a modeling aspect going on we're saying that effectively organisms are sort of in a process of approximate bayesian modeling of their the causes of their sensory stimuli then there's a kind of meta level of analysis where we can actually as scientists write down a generative model that says well the markov blanket is here and then score that model in terms of the variational free energy and use a variational free energy approach to kind of find where the the boundary

is of the variational free energy minimizing system so i i kind of take that uh you know meta aspect to be a strength of the the framework nice agreed um we'll go to stephen and then mel yeah and so following on from that as the the the piece that sometimes gets maybe lost here but is in the other papers when you look at regime of attention is that it's not just that the system cleaves like the whole the the the mark of blanket gives a capacity to change how you take your from the stream that are coming in like i suddenly feel my hand then i feel my face and i suppose one of the things that would be an example as well of something that's a very large scale is the threat of climate change and i'm thinking about when i start thinking a glacier in antarctica starting to come away and go into the the water and possibly raising sea levels in 20 years my physiology changes and my whole you know when i bring that into my considerations the way i take my attention and the way i tune to a lot of things around me in the world changes so you know is it is that has my system boundary changed or i would say that it's more about my attention or integration that's changed so that's where i think there's a question around whether the word systems could be overused sometimes and it's maybe sustained with this kind of statistical dynamic is quite quite useful right it's uh statistically influencing you whether you like it or not and then your awareness and your attentional regime is something that's almost separate mel oh i just wanted to say um goodbye because i'm i'm supposed to be in another girl right now thank you for participating now awesome thanks for having us this is this is lovely cool so in the last couple of minutes i just wanted to get all the figures on the screen just so we could run through them let people take the next week to digest them a little bit more fully come for whatever exciting parts they were so when i look at figure one i just see on the top the simplest framing of system from surrounding and if you think that that blurry that's a blurry boundary then you're not talking about a system so to be able to define something is to uh you know to delimit is to define so it might not be the most functional boundary but you can start there and then you ask well what kinds of states or interventions connect the system to the environment just like muddy was describing and those are described in this model as and if we had to put english words on them they would be sensory states and active states but ultimately it's really about what the model actually says figure 2a goes into a bit more detail in that action perception internal external two-stroke engine as i think about it this feedback loop from internal and external states connected through sense and action and then the markov blanket is what is being pierced bi-directionally by sense and action and as we've talked about before and in some of the work on niche construction one systems internal is another system's external and so in some sense what is

being uh discussed is that blanket state because there's some symmetry between the internal and the external in that both internal and external states have their own self-causing or autonomous dynamics as well as influence each other through sense and action respectively so figure two is just showing that again in the context of a human and then also connecting it a little bit more closely to the free energy of energy minus entropy and so again another day for talking about the math figure 2b just talks about how those formalisms of free energy respectively complexity minus accuracy on the action bound and then divergence plus surprise on the perception bound how the math sort of sketches out another day figure three is an illustration of operational closure so what they note here is that the black circles are forming part of an operationally closed network of self-organizing processes and this is something like teleological like if you're talking about a sustainable ecosystem you don't want the sustainable ecosystem to be defined from your front door to your trash can you're talking about operational functional closure existing at a higher level of analysis so that means the multi-level perspective is going to be important but at one level of analysis or at multiple we can ask has operational closure been achieved have we done enough system diagramming or mapping of causal relationships or dependencies such that we actually have the system's dynamics separated out from dynamics outside of the system if i can just jump in the the motivation for including this figure uh is part of the argument that we're making i think that's controversial is that basically most of what the inactive approach does uh we can do with the active inference approach but in a way that isn't just uh like uh well in a way that's formally backed you know by like a principled uh physics-based approach uh and so we're just discussing here the notion of operational closure uh that inactivists use to define an autopoietic uh entity like a self-producing entity and the argument that we make immediately after is that well uh active inference doesn't just doesn't just say there is this operational closure it tells you why there is operational closure nice and also to reference what we're talking about with mel just briefly um kant performed a teleological analysis and said that the organism is not just the limits per se of the epithelium not a physical boundary but it's actually the teleologically closed entity and so that is really consonant with the con with the idea that the eusocial insect colony is an organism not a superorganism by calling it a super organism we reify the boundary of organism being at the epithelia and people say oh well it's like a superorganism it's all these little insect organisms working together but actually because of the major evolutionary shift in eusociality in the ants for the colony is the organism we don't draw a little single cell organism barrier around each one of our cells and then spend all of our time running the math on the

altruism between the nerve cell and the muscle cell and the the sex cell and things like that we just say no that's the teleological level the liver is going to make sense in a broader end and that end is the and then the question about what the privileged level of analysis should be the organismal teleological analysis says right the liver and the stomach makes sense at the level of the organism but in some sense doesn't the organism only make sense in the context of its cultural niche and of its social relationship so why stop there where should the teleological closure end where should the functional closure end and that's what demands i think a true multi-level perspective because there's going to be questions where the functional boundary the epithelium around the person or around the single ant nest mate but there's other times where what we're talking about is the closure around the organism as the teleological hole and that's not always the same thing as the epithelium yeah richard yeah i i think this general argument is is really captured well in the in the craver and danted mechanistic philosophical perspective to where a phenomenon is defined by the entities and activities that generate it or produce it and so whatever phenomenon you just you define whether it's the ant it's the colony there's going to be some mechanism that that generates it and that mechanism is decomposable down to whatever you want so the colony can be decomposed down to the individual ant which can be decomposed down to the various biological structures and really when you want to move up and down you're just simply identifying okay what are the entities and activities that generate a phenomenon that then interacts with other phenomenon entities and activities to create that right and you build up and down because i think i think it gets really dangerous in sociology we do this a lot to where there is no individual there's only society and there's theological reasoning where it gets very hard to keep track of okay well bob commits a crime not the entire society that bob's a part of so how can i drill down on bob as an entity that i want to talk about but then recognize that bob is part of a of a larger structure and not reducing it to the the group as the entity in itself yes it's 100 um related to that book of helen longino whether you know was it the arm that committed the aggression was it the person or was it the group and so it's funny i'm sorry you mentioned that because i mean that's kind of a symptom in schizophrenia where a lot of criminals will say i didn't kill him the bullet did and and then logically that makes sense in terms of that reasoning but any rational person would say well no you pulled the trigger so therefore the bullet is responsible you know are you responsible for the bullet's pathway well no no i didn't kill him the bullet did so that that logic i think when pushed to its logical extreme um breaks down pretty quickly but yet nonetheless i think we do tend to implicitly rely on it a lot in our yep and before we go to stephen it's again it's those

causal stories guns don't kill people people do well depending on cognitive diversity that statement is going to sound true or not to you and someone could say well it's not even the bullets it's actually blood loss or it's not even blood loss it's reduction in blood pressure where does the causal chain end and so at some point we need to cut through the descriptive approach with an action-oriented approach okay are we gonna ban bullets guns people where do we stop there's no single answer just from identifying the territory to action and so these are really the big questions about causal relationships in multi-level systems we'll go with stephen and then muddy yeah i think the ability to have these is made possible because of processes so we think about this in art because art is seen as a process that is creating something so in the same way with the ants there may be a mechanism by which ants chew up and masticate pieces of um wood or mud and there's a mechanism by which they build the ant colony and that ant hill has a structural integrity that allows air to flow through certain systems of movement that creates a cooling effect however the processes which i think the markov blankets are working with are what make the mechanisms possible to happen like the ant is following these kind of statistical dependent processes to do what it does and in doing so it has a mechanism just like if someone's firing the bullet there's a process the body's using to infer yet the mechanism is something we see so i think this is these these things kind of entwined but they are slightly different agreed muddy and then back to maxwell um i guess this is where my original gripe with philosophy of science came is from that from an empiricist point of view for me you know where do you snip the causal chain is something that can be empirically defined you know if i have an extremely strong coupling between action a and action b such that action b doesn't occur without action a then move the markup boundary beyond it right that's that's good so any time anyone says guns uh you know bullets don't kill people um people do or whatever sorry people don't kill people bullets do i don't know what the the spurious phrase is or whatever but you can say well let me actually empirically look at this what percentage of uh and i'm sure there is like somebody's going to come with an example oh there was this gun that went off itself whatever but statistically speaking if i threshold that i can find basically a probability of one or very very or tending towards this that the gun was fired by a human being so i think it's a bit of a when somebody when somebody presents such a philosophical argument i'm like well you know there is an empirical response the science gives you these empirical tools to infer these sorts of things so that's kind of where my my original gripe about the phosphorus science coming out again i've softened on it but i just wanted to say i think there's an answer to these things philosophically and it comes from

sampling the data cool uh maxwell before we finish out the figures cool uh well i just want to think about something that was mentioned in the chat by richard is that the so there isn't any i think systematic work that relates the free energy stuff to uh neo-mechanistic philosophy a la craver something i'd like to discuss with you all next week is um so as i understand it the multi-level framework i started off as a neo-mechanist i have to say like uh you know the the craver bechtel orientation was really something that i was pursuing and around 2013 2014 i i kind of spent a lot of time trying to see whether i was able to you know use the mechan the neo-mechanistic framework to articulate multi-level causation and my big gripe with uh neo-mechanism at least in its most popular renditions is that it effectively eliminates uh cross-scale uh causal effects so there's a part of this that i buy which is that you know part of my rejection of metaphysical levels comes from the neo-mechanist approach uh but i i think that because with the the the story we should discuss this maybe in in another podcast but the their story is something like uh what looks like uh you know inter-level causation is really uh is really just uh it's it's just the same causal stream that you're redescribing at another level one of the things i like about uh the fep formalism is that while it's consonant with the broad outline of uh neo-mechanistic philosophy it does explain the way that you have causal effects that trickle up and down um so i think that's like an outstanding issue that that should potentially be uh discussed in a sort of flag that for the next episode yeah good call um alex k and then i'll close out the figures all right i was i was actually gonna lower my hand because that maxwell said we could discuss this another time but i guess i briefly just wanted to say yeah i think if you if you go back to like kant's definition of mechanism which i can't recite but uh basically he contrasted mechanism with systems that involve top-down sort of causation as well like you know that holes can affect parts i i would guess you could still get some kind of mechanistic description at each scale or like approximately or something but anyway i think it's an interesting topic oh yeah yeah just to be clear my view is that the fep is broadly consonant with uh and is in fact an exemplar of the neo-mechanistic research strategy as outlined by bechtel and craver and so on but i think it it does impose a different metaphysics of levels uh than the one that's been uh attempted at least in a few papers uh that i saw by bechtel and company like the one called looking up down and around i think uh where they essentially say that well there's no such thing as cross-scale uh influences they're really just big things interacting with small things and uh that's where i i think i strongly disagree uh for reasons that hopefully will unpack uh next week and that are related to how these multi-scale systems are actually constructed and i mean essentially you've got the same base system uh that's acting so it's like i don't know it's a

ladyman and ross style there are only basically quantum fluctuations or whatever but what you're doing is whenever you define a new scale of interactions you're adding a slower and spatially larger set of constraints on the same fundamental set of things so there is something like cross-scale talk in the sense that what it is to be a multi-scale system is uh essentially to have components that are constrained by successively higher level uh constraints on the dynamics uh so but i don't think ultimately this this conflicts with the philosophical perspective of neo-mechanism which ultimately at the end of the day just says well there are parts and processes and a mechanism is a coordination and an organization of parts and processes that produces some kind of interesting output that we can study so cool yep these are um alejandra go ahead yeah yeah thank you and just following up this idea uh you're saying max well i i was thinking if is there any differences in how these hierarchies um in terms of blankets uh work i was wondering um at the cellular level uh maybe belief dating is kind of risky so the cell has to behave like having this generative model but um [Music] it's it's like it's phenotype but it's not learning uh i don't i don't know i'm maybe i i don't have um very clear what i'm thinking but if you go like up to the scale thinking like in the brain um level the blanket now um belief of dating it's it's it's crucial for for um like this bodily interactions and effective bodily interactions and this coupling with the environment so um my question is is there any differences in terms of how these micro blankets and blankets of blankets blankets work um and this it's also a hierarchy and how this hierarchy belief updating works let's let's return to this great question when we have a lot more time next week which is about the similarities and differences with different systems it's really something we should consider because i was gonna say this is like the question yeah that we address in this paper in large part uh i mean a quick answer to your question is that formally speaking there's a kind of self-similar pattern that that is occurring across all these different scales right so you have a non-equilibrium steady state density and a generative slash generative model uh the the and that is uh kind of directing policy selection in a way that maintains the blanket but that's a very very uh like abstract or information theoretic description and in effect uh like basically the the kind of data that's involved at every scale the mechanisms involved to make sure that your data the data that you're sensing fits your preferred data distributions all these things are going to be radically different from one scale to the next uh i mean yeah clearly the way that we update beliefs about ourselves by having a conversation while it has a formal while it may formally be identical on a kind of process or information flow level uh to like what our cells are doing like clearly this is different right like the the data streams with which we're dealing are

different the ways that they're being managed they're different uh so yeah i mean the the there is a kind of fractal self-similar structure uh but it's at the level of the information flows not so much what makes up uh so in mechanistic terms the parts are different but the processes might be yep and it will be awesome to go into more detail because there's clearly differences between cells that embody their beliefs through morphology that might be slower time changing and brains that can rapidly emulate or at least you know simulate various phenotypes being neural states and in a previous paper we talked about whether those neural states are themselves phenotypes just to quickly run through these last figures figure four is an example of auto catalytic closure so this could be like a metabolic network or self-replicating nucleotide sequences or programs that replicate themselves with or without mutation and the point here is really to say that that kind of causal dependency structure that we saw in figure 3 it might not just be bumper cars or ants bumping into each other it could be something like different types of information that are connected to each other and so as we start to think about different spatial temporal scales and all these different mechanisms that exist we want to find those invariances at a structural and dynamical levels that help us study this diversity of systems using a common formal framework figure five is something we already looked a little bit at but it is this fractal idea zooming in on the fractal broccoli and basically the left side has uh sensory and action states surrounding three gray nodes but that's itself zoomed in from a swarm similar agents and so in a tissue maybe it's very patterned like in the skin cells the boundaries between the cells are very stable so they can have adhesive molecules that make it a functional skin sheet and they're still exchanging information in that grid topology but also maybe the swarm on the right side consists of moving subunits so like steven was talking about with a swarm and at that point especially it is important to have a statistically driven understanding of causal relationships rather than one that's just around drawing a dashed line around the system as we visually see it and then just lastly figure six which i think will be perfect to save for later especially because it in some ways does connect this action inference generalization to specifically morphogenesis which is an example of a phenotype decision making that people might not always expect to be put in the same category along with what it is our brains or cpus are doing but i think that's a perfect jumping off point for next week when we uh yeah nice and going through this is going to hopefully address alejandro's question uh i think so yeah this is a good jumping off point i think perfect and we have a few questions just prepared for next week but we're not gonna go to them now we will talk about that next time that concludes the main section and we'll also just have anyone can raise their hand and give a

final point but i just wanted to say that we'll provide a follow-up form to our live participants i'll put it in this jitsi chat as soon as someone starts to give a final thought any feedback suggestions or questions our listeners participants are always welcome of course everyone's engagement is welcomed and people can stay in communication with us so if anyone has their hand raised i'll be happy to hear their first thoughts while i post the link nobody's hand is raised but thanks again for the continued engagement with our work um it means a lot and i think this is extremely useful uh like both i think for for us as a little community working through these things and uh you know for the broader uh audience the free energy active inference stuff so i'm very grateful thank you very much awesome um muddy um raised your hand and if anyone else wants to give any final thoughts what they reflected on from today's discussion what they'd be excited to hear about in next week discussion uh i was just going to say um thanks sir this is really really great i think um first of all just on a personal level it's great to get back involved with people who are much cleverer than i am and be surrounded by this this new knowledge but secondly i can really start to sense the snowball effect in the active inference community it's only a matter of time before it really leads over into industry it's a really really exciting time to get involved in the space thank you all that's really good cool richard and anyone else freely raise your hand yeah i i think this is really interesting and i think there's there's so many different points of connection and interaction with so many different disciplines and um i think one of the difficulties is the barrier to entry is just so high in terms of the terminology the mathematics to try and rapid your head around these things but once you begin to understand it conceptually i think people can start go okay i know what this is and i can talk about this and so i think one thing to maybe you know discuss moving forward is are there ways to kind of get almost sandbox working groups where people can kind of start really empirically playing around with ideas and data sets um that kind of emerge naturally out of these conversations because you know cold calling somebody and saying hey i've got this idea can you explain to me what a marcotte blanket is let's work together people are like no son i got things to do i'm sorry i think you know there's a way that we can kind of merge from these really great productive working our discussions into maybe something where people start producing actual collaborations across disciplines i think would be a really fantastic thing awesome i i just want to reiterate also that there's this reading group uh that's ongoing on the carl firstens monograph so for those of you who are interested in actually uh you know uh surmounting the barrier to entry as it were uh we're we we've recorded uh the the chats and everything so if you do want to join just uh drop me a dm um and yeah we've grown

to this is originally going to be like me and eight buddies just reading through the monograph and we've grown to a community of almost 100 scholars uh yeah following this uh like remotely with the recordings available and we're going to turn these recordings heavily edited and everything into pedagogical material a kind of like video based a companion to a free energy principle for a particular physics uh so yeah if anyone is interested uh the group is kind of semi loosely open um to especially the to people who are doing active research uh in the tradition so uh yeah drop me a dm and i'll all hook you up cool stephen and then anyone else if their last thought comes to mind yeah well just to reiterate yeah it's really useful i like to see everyone nibbling at this from different directions are sort of it it shows a lot the questions everyone's asking it's like because otherwise you're thinking it's just me that's thinking this and sometimes it is and then sometimes it's not so it's kind of good to know which is which because sometimes it's not clear it's just a figure of my imagination as the only other thing i thought might be cool to bring is that new paper that came out on reciprocal active influence that's just come out might be interesting at some point to think about because that might open up some another level of development from this paper a very interesting paper by matt sims uh at edinburgh who um i'm i've been very excited about this stuff so matt sims developed a um an account of symbiosis that draws on the multi-scale integration framework that we've uh developed uh where he talks about bi-directional uh multi-scale integration so not just integration of the parts of of one biological organism but the integration of effectively a symbiotic union of organisms with different phenotypes uh the paper uses the uh i think it's the bobtail squid that uh gets colonized by bacteria and yeah it's a very very interesting paper and i highly recommend it cool cool very interesting sasha yeah i just wanted to reiterate what stephen just mentioned um it's great to have this ongoing and to hear how different people phrase the questions that i think we're all thinking so i'm really looking forward to next week's discussion um and kind of uh seated with the question that alejandra stated earlier perfect well thank you all for participating the google form for you to provide any feedback for the live participants is in the chat for people who are watching this live or in replay please provide us your feedback so we can make this better on-ramp for active inference the community the ideas and ultimately the projects too so it's great stuff another one in the books 5.1 for active and i'm going to terminate the live stream thanks so much for listening